Auteur: Seda Jusjajeva
Studierichting: Informatica
Oefeningenreeks 6A nr. 9.6

$bc - ad = 104 - 54$

$\Leftrightarrow (x_1 + v) \cdot (x_1 + 2v) - x_1 \cdot (x_1 + 3v) = 50$

$\Leftrightarrow x_1^2 + 3x_1v + 2v^2 - x_1^2 - 3x_1v = 50$

$\Leftrightarrow v^2 = \dfrac{50}{2}$

$\Leftrightarrow v = -5 \vee v = 5$

$v = -5$:

$x_1 \cdot (x_1 + 3(-5)) = 54$

$\Leftrightarrow x_1^2 - 15x_1 - 54 = 0$

$\Delta = (-15)^2 - 4 \cdot (-54) = 441$

$\Rightarrow x_1 = \dfrac{15 - 21}{2} = -3 \vee x_1 = \dfrac{15 + 21}{2} = 18$

$x_1 = -3$:

$x_2 = -3 + (-5) = -8$

$x_3 = -3 + (-10) = -13$

$x_4 = -3 + (-15) = -18$

$\Rightarrow -3, -8, -13, -18$

$x_1 = 18$:

$x_2 = 18 + (-5) = 13$

$x_3 = 18 + (-10) = 8$

$x_4 = 18 + (-15) = 3$

$\Rightarrow 18, 13, 8, 3$

$v = 5$:

$x_1 \cdot (x_1 + 3 \cdot 5) = 54$

$\Leftrightarrow x_1^2 + 15x_1 - 54 = 0$

$\Delta = 15^2 - 4 \cdot (-54) = 441$

$\Rightarrow x_1 = \dfrac{-15-21}{2} = -18 \lor x_1 = \dfrac{-15+21}{2} = 3$

$x_1 = -18$:

$x_2 = -18 + 5 = -13$

$x_3 = -18 + 10 = -8$

$x_4 = -18 + 15 = -3$

$\Rightarrow -18, -13, -8, -3$

$x_1 = 3$:

$x_2 = 3 + 5 = 8$

$x_3 = 3 + 10 = 13$

$x_4 = 3 + 15 = 18$

$\Rightarrow 3, 8, 13, 18$

# References